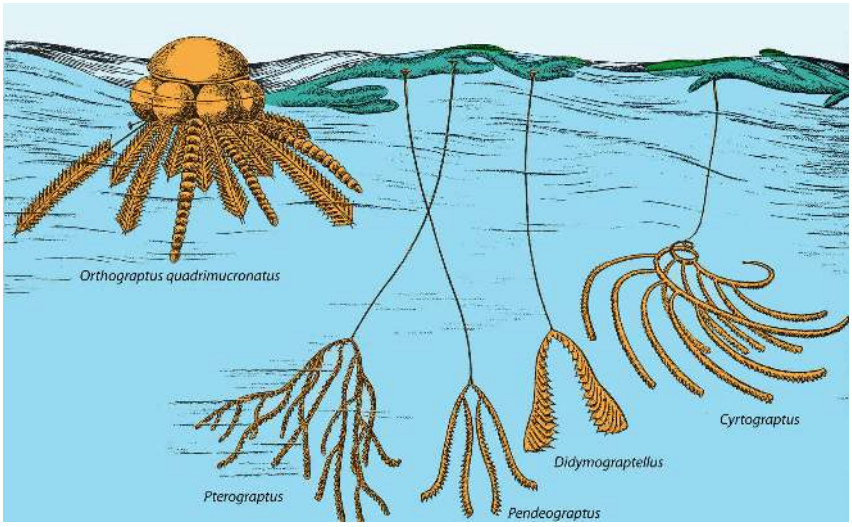
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Macrofossils: Graptolites and Conclusion

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Macrofossils: Graptolites

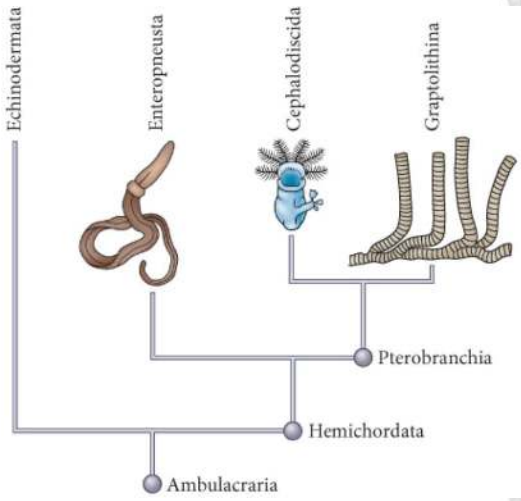
Russell Garwood | russell.garwood@manchester.ac.uk

Image source: Maletz, J., 2015. Graptolite reconstructions and interpretations. Paläontologische Zeitschrift, 89(3), pp.271-286.

Introduction and biology

What is a hemichordate?





Hemichordata: A phylum of bilaterally symmetrical soft-bodied marine invertebrates that comprises two classes: the Enteropneusta (acorn worms), which are cylindrical burrowing animals; and the Pterobranchia, which are colonial forms with vase-shaped bodies.

Left: Tassia, M.G., Cannon, J.T., Konikoff, C.E., Shenkar, N., Halanych, K.M. and Swalla, B.J., 2016. The global diversity of Hemichordata. PloS one, 11(10), p.e0162564.
Right: Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons.

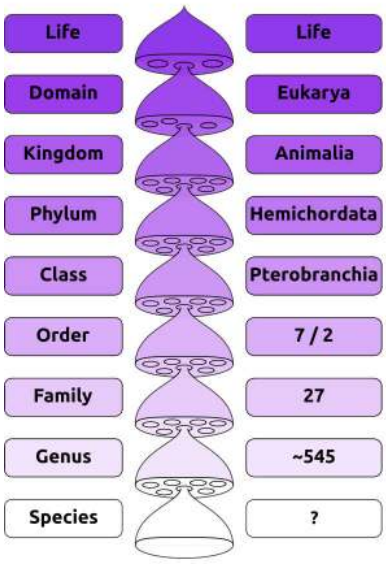
What is a graptolite?

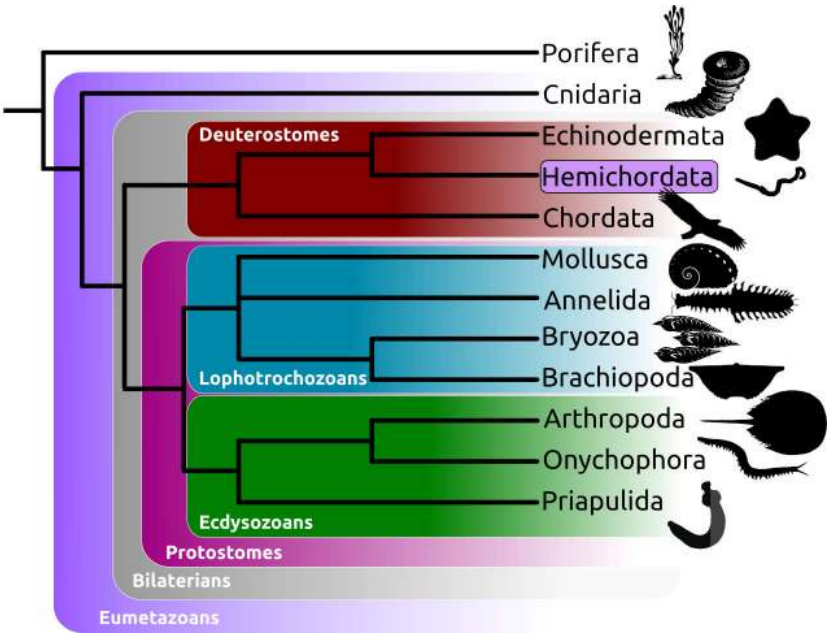




Graptolite: A member of a group of extinct marine colonial animals that were common in the Palaeozoic era. Graptolites are generally regarded as colonial hemichordates.
Pterobranch: Class of minute, fixed, colonial, deep-sea organisms which secrete an external cuticular skeleton in which they are housed (e.g. Rhabdopleura).
Photographs by James St John.

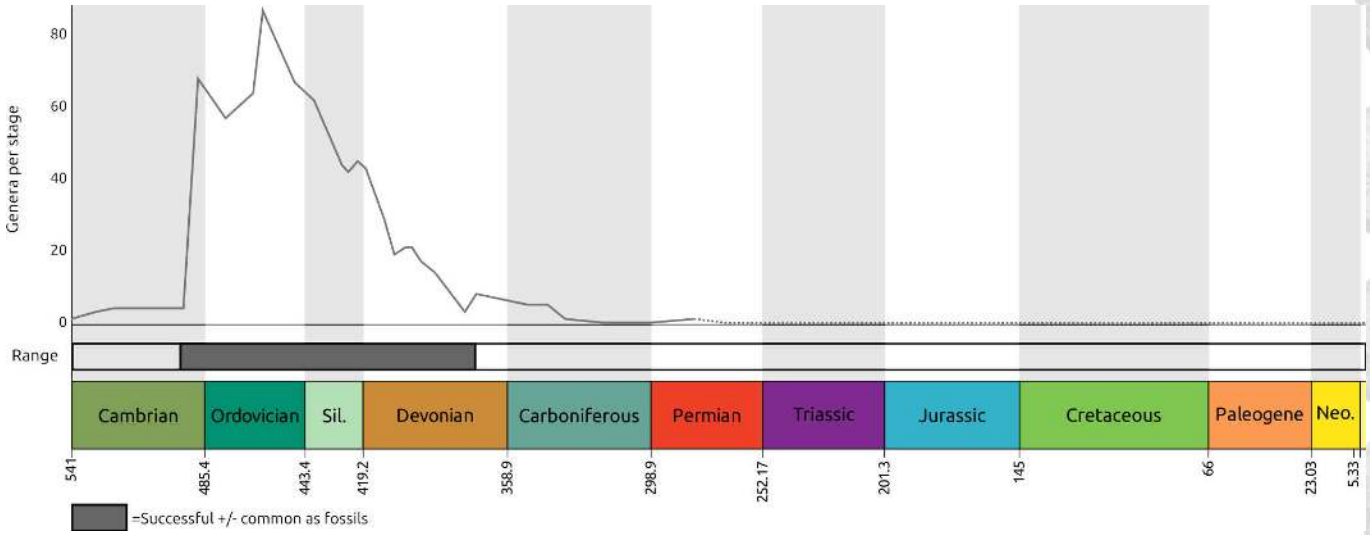
Where do they sit on the tree of life?



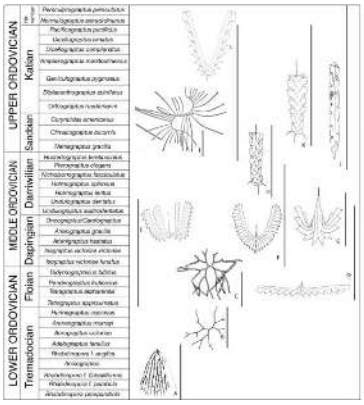


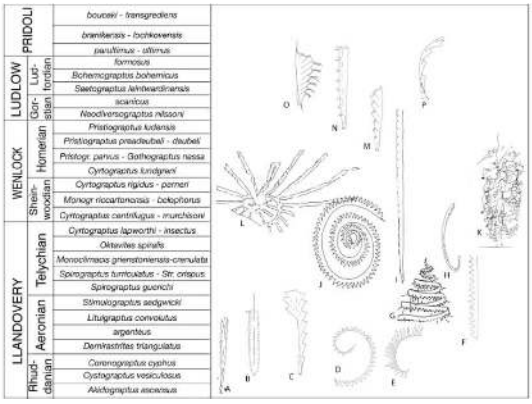
Maletz, J., 2014. The classification of the Pterobranchia (Cephalodiscida and Graptolithina). Bulletin of Geosciences, 89(3).

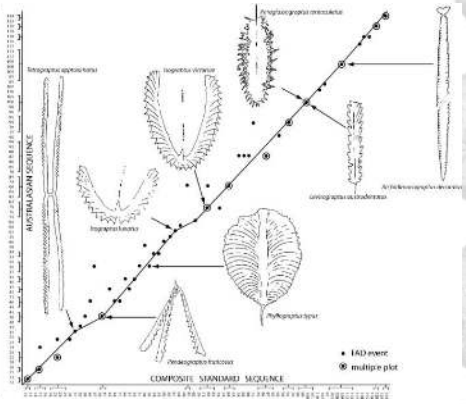
How are they distributed through time?



Fossil data from PBDB.

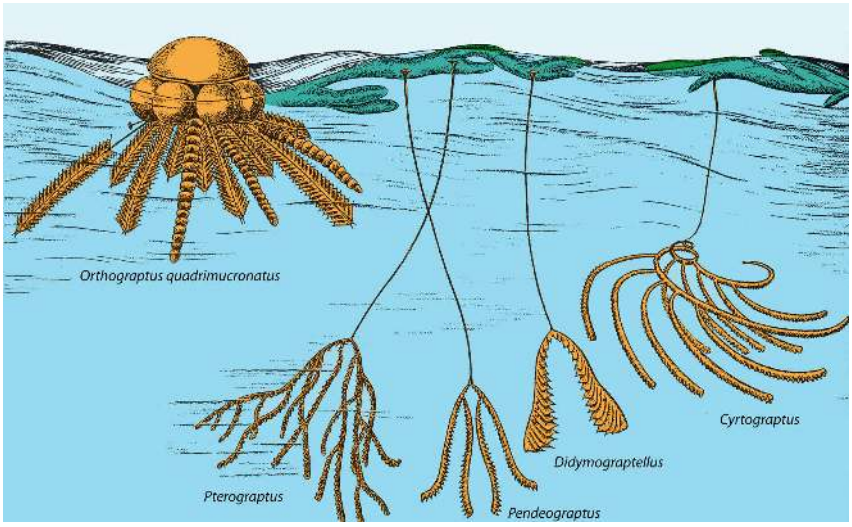






Source: Maletz, J., 2017. Graptolite paleobiology. John Wiley & Sons.

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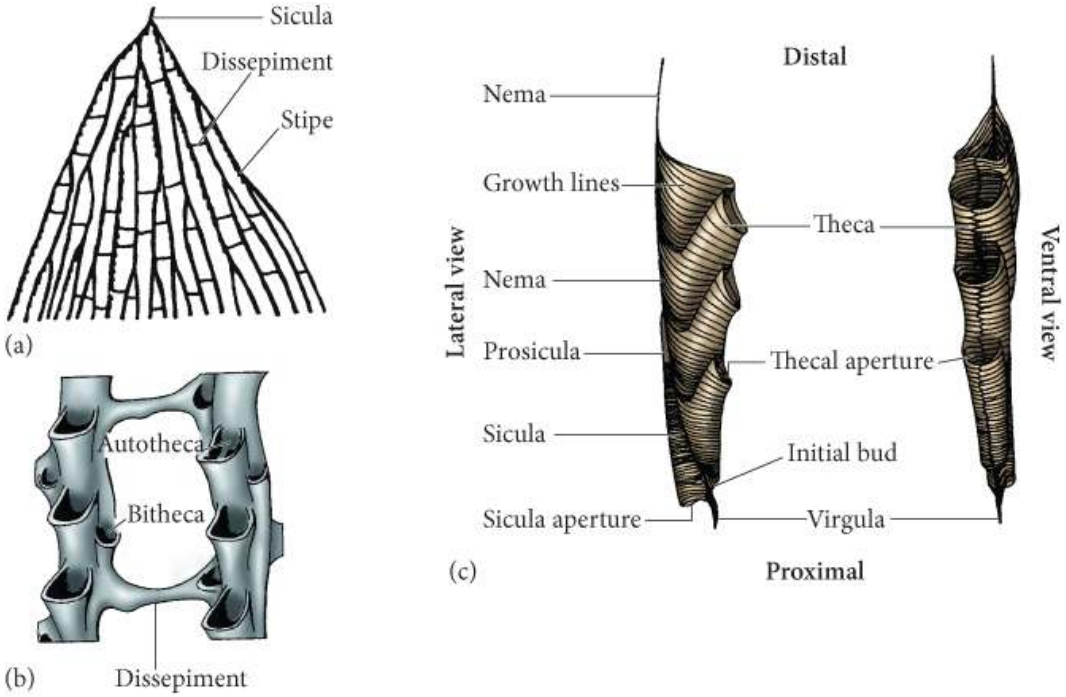


Macrofossils: Graptolites

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Image source: Maletz, J., 2015. Graptolite reconstructions and interpretations. Paläontologische Zeitschrift, 89(3), pp.271-286.

Morphology / Anatomy



Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons.

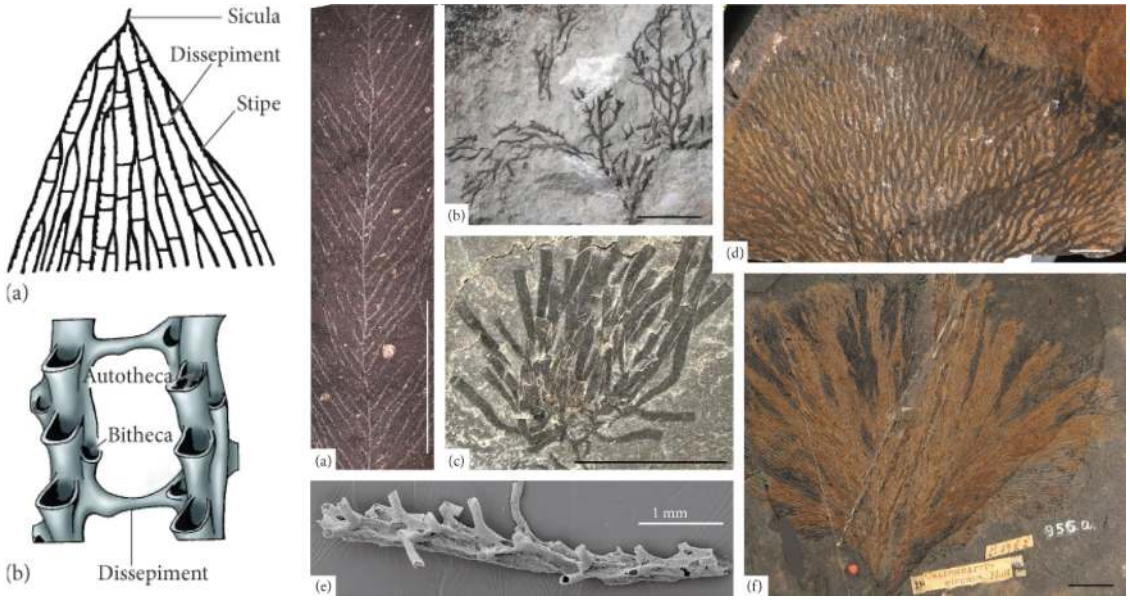
Zooid: In colonial invertebrates, an individual member of a colony.

Sicula: The skeleton of the initial zooid of a graptolite (Graptolithina) colony.

Stipes: A branch of a rhabdosome (graptolite colony). A stipe or stipes originate(s) from an initial conical cup (sicula). The number of stipes can range from one to as many as 64.

Theca: In graptolites (Graptolithina), one of the cup-like structures that occur along the stipe, each of which houses an individual member of the colony.

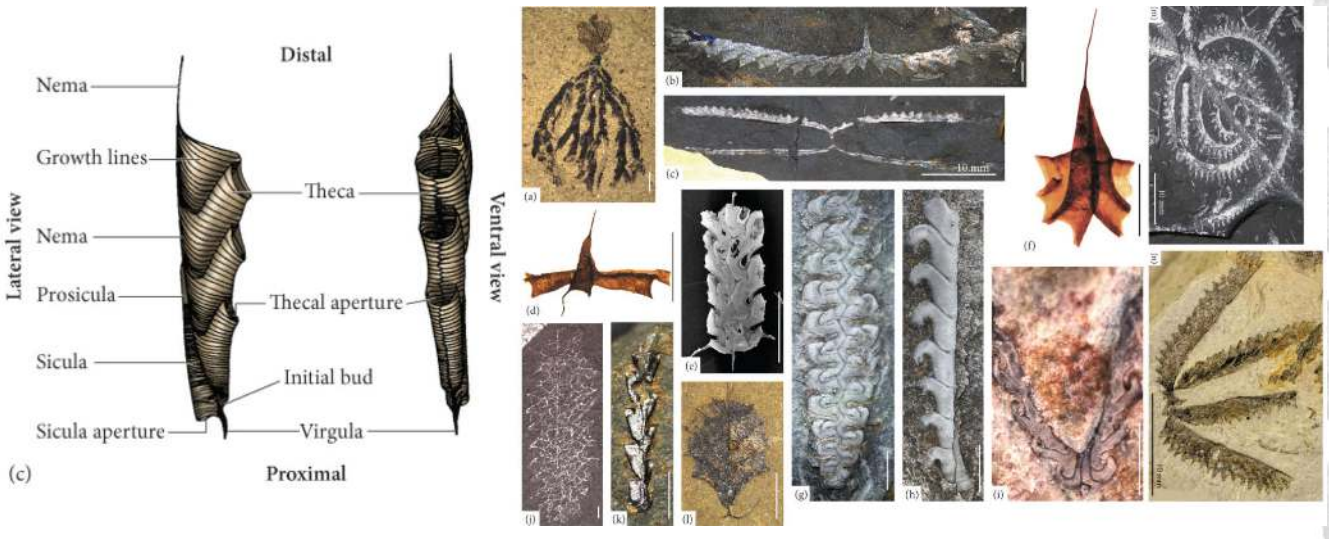
What do dendroid graptolites look like?



Dendroidea: An order of graptolites that existed from the Middle Cambrian to the Lower Carboniferous. They are many-branched, with numerous small thecae, in some connected by struts.

Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons.

What do graptoloid graptolites look like?



Graptoloid:An order of graptolites that existed from the Lower Ordovician to Lower Devonian.
Nema: In graptolites, a long, hollow, rod- or thread-like structure extending beyond the sicula
Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons.

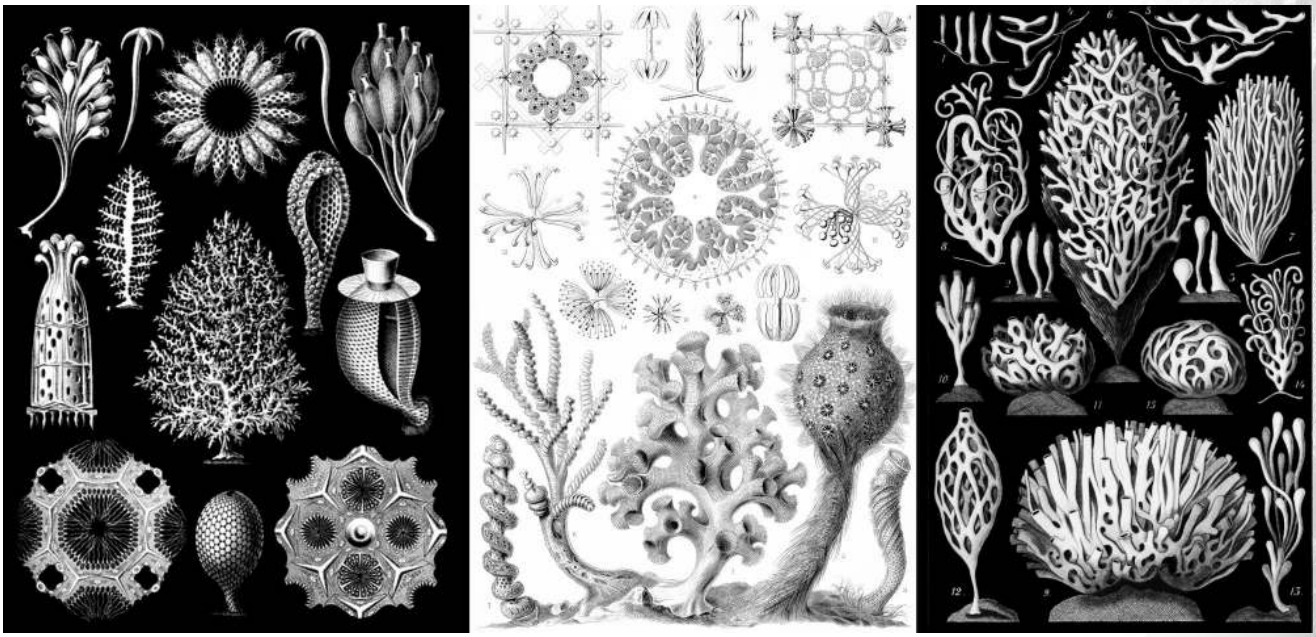
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Macrofossils: What we have missed

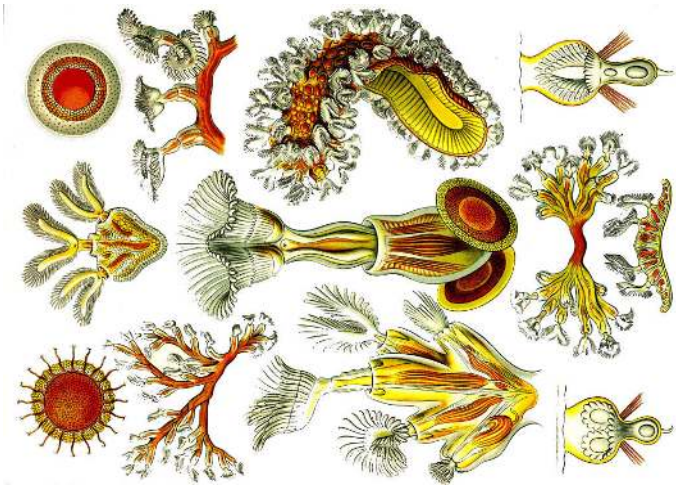
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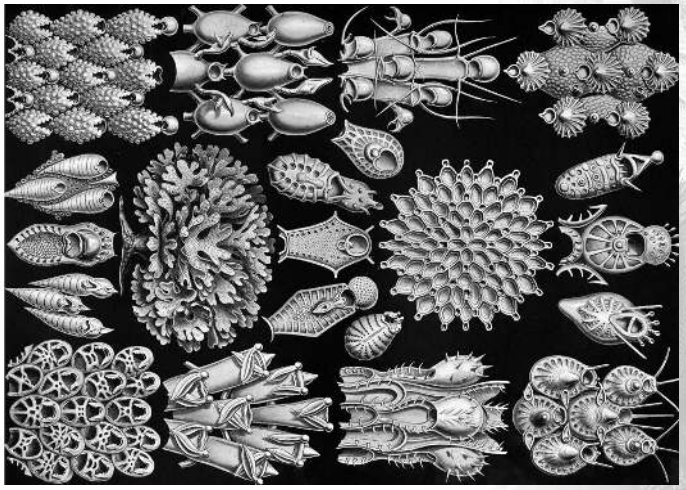
What we have missed: Sponges



Sponges: A phylum of multicellular animals. Sponges are sessile, benthic, filter feeders with a bag-like body.

What we have missed: Bryozoa





Bryozoa: Phylum of small, aquatic, colonial animals, related to the Brachiopoda

What we have missed: Annelids





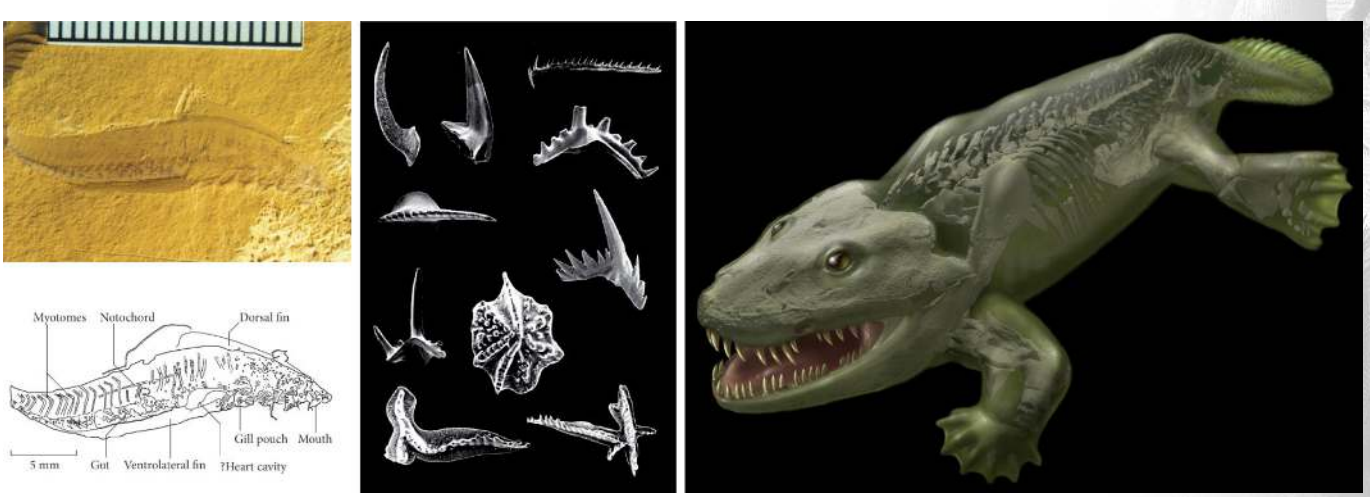
Annelida: Phylum of coelomate worms which possess a definite head and good metameric segmentation

Serpulid: A small marine annelid worm which lives in a twisted shell-like tube.

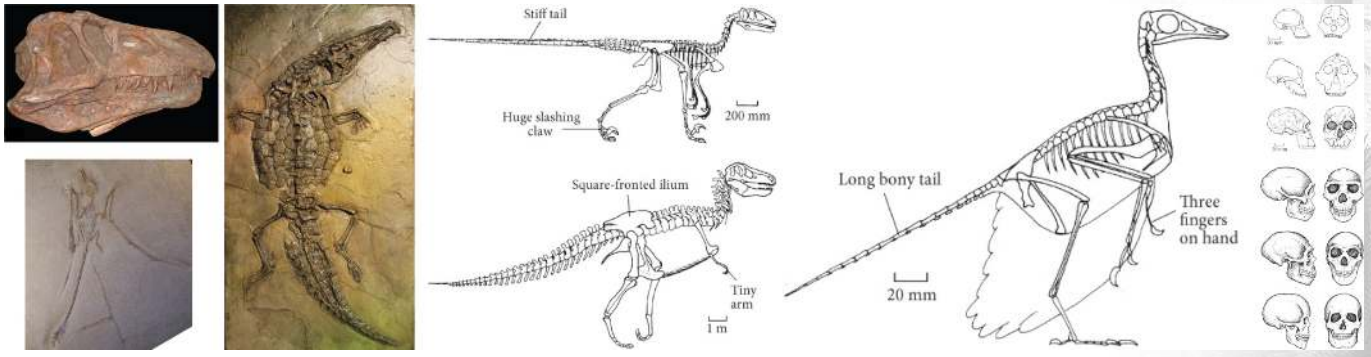
What we have missed: Vertebrates



Hackel, E. 1899. Kunstformen der Natur.



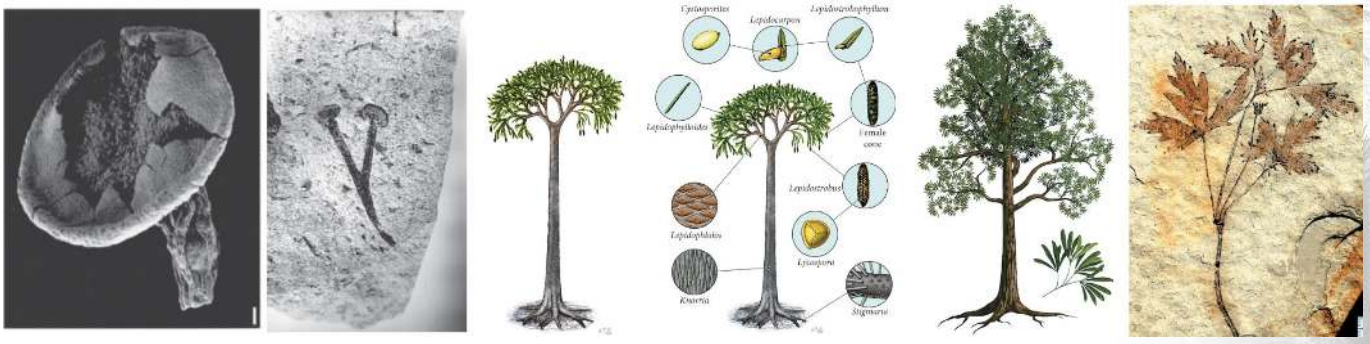
Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons.



Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons.

What we have missed: Plants



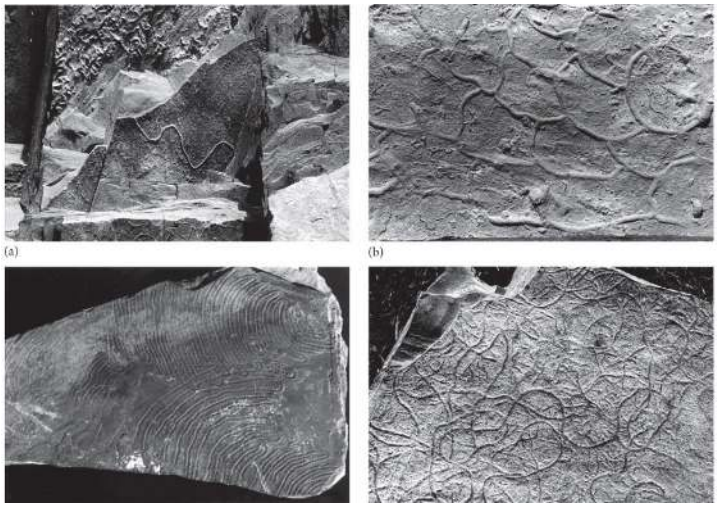


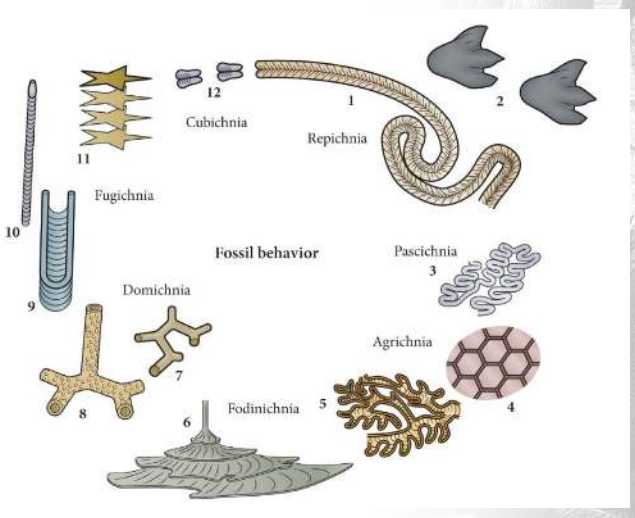
Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons.

What we have missed: Trace fossils



More: Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons – Chapter 20. | All photos by Russell





Source: Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons – Chapter 20.

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Macrofossils as indicators of the environment

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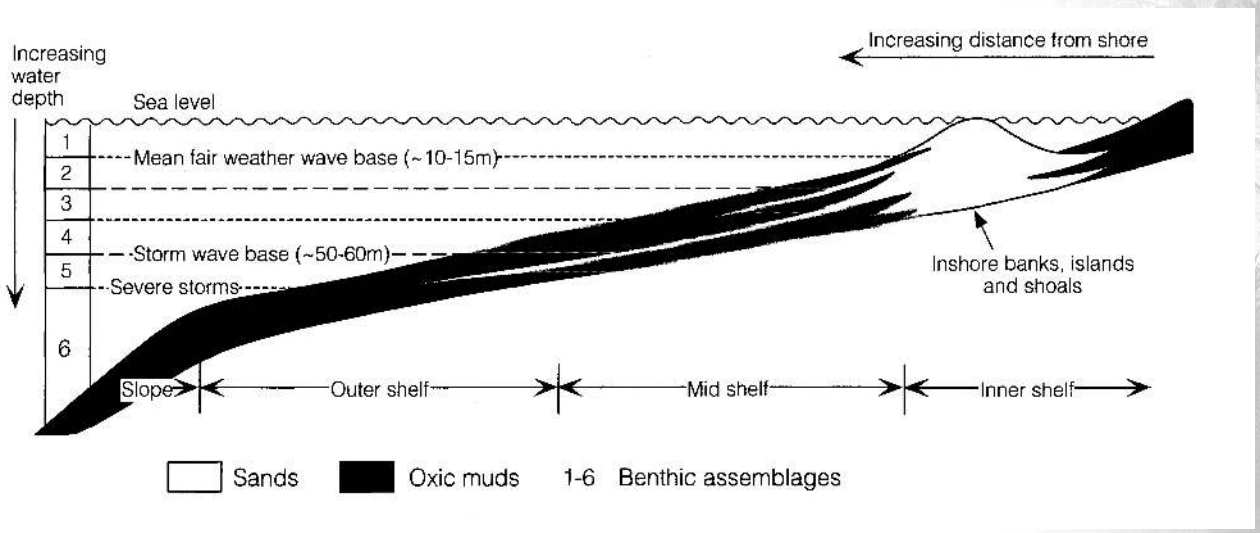
Why do we need this?





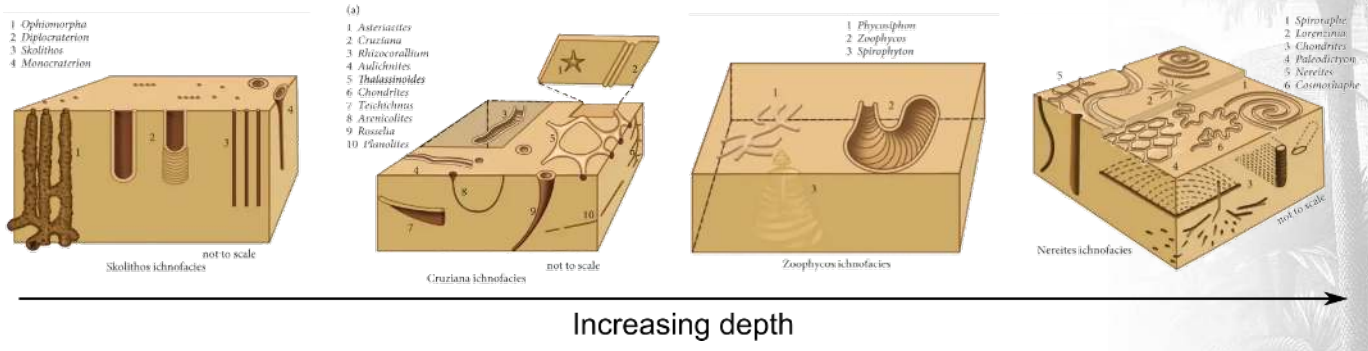


Example: Discriminating environments on clastic shelves...



Biofacies: Rock unit, or an association of rock units, characterized by the presence of a fossil assemblage that is restricted to that particular facies and that is typical of a specific environment.

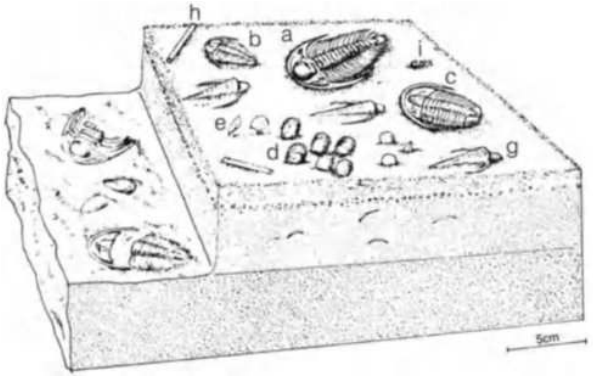
Definition source: Allaby, M., 2020. A dictionary of geology and earth sciences (5th ed). Oxford University Press. Image source: Brenchley, P.J. and Harper, D., 1998. Palaeoecology: Ecosystems, environments and evolution. CRC Press.

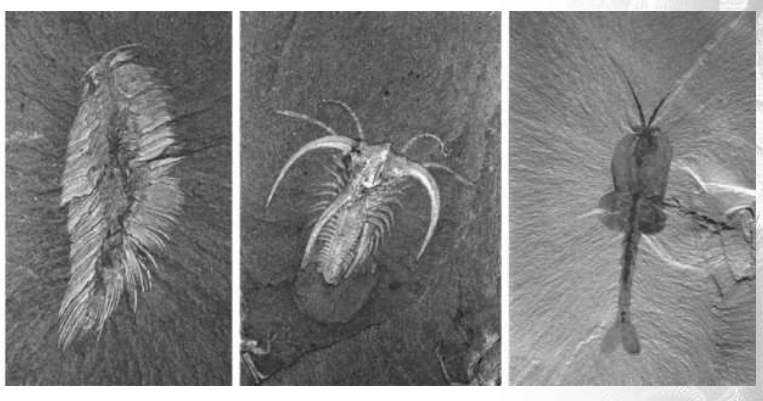


Ichnofacies: A rock or sequence of rocks characterized by its lithology , inorganic sedimentary structures , and specific trace fossils. The traces represent the behaviour of fossil organisms under particular environmental conditions.

Definition source: Allaby, M., 2020. A dictionary of geology and earth sciences (5th ed). Oxford University Press. Image source: Source: Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons.

... So let's add time.

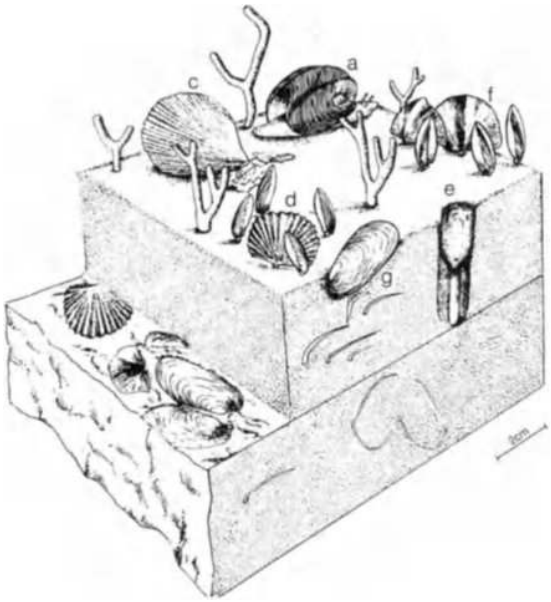




Palaeozoic

Left: Typical low-diversity, shallow marine Cambrian community, a. Paradoxides (Trilobita), b. Bailiella (Trilobita), c. Solenopleura (Trilobita), d. Lingulella (phosphatic brachiopod), e. Billingsella (billingsellide brachiopod), f. Micromitra (phosphatic brachiopod), g. Hyolithes (Caliptoptomatida), h. Hyolithella, (uncertain affinities), i. agnostid (Trilobita).

Left: Brenchley, P.J., Brenchley, P. and Harper, D., 1998. Palaeoecology: Ecosystems, environments and evolution. CRC Press.
Right: Smithsonian Institution.





Palaeozoic

Left: A shallow marine (BA2) upper Ordovician community, showing a bivalve-brachiopod-bryozoan association.
Right: An outer shelf (BA4 to BAS) Upper Ordovician community with a high diversity brachiopod-trilobite association.

Brenchley, P.J., Brenchley, P. and Harper, D., 1998. Palaeoecology: Ecosystems, environments and evolution. CRC Press.







Mesozoic to Recent

Left: A shelly lime mud community from the middle Jurassic, belonging to a sheltered shallow marine environment with a varied infaunal and epifaunal bivalve fauna.

Brenchley, P.J., Brenchley, P. and Harper, D., 1998. Palaeoecology: Ecosystems, environments and evolution. CRC Press.
Vermeij, G.J., 1987. Evolution and escalation: an ecological history of life. Princeton University Press.
Right: Jurassic World, Universal Pictures.